

Parametric Inference

Two-Day Exam Revision Handout

Built from the 10 lecture notes (29 Jul – 28 Aug 2026) and Homeworks 1–3

0. The two-day plan

Your position: Lectures 1–5 (estimation) were studied but are stale; Lectures 6–10 (Bayes, minimax, testing, asymptotics) are the new material. The homeworks tell you the exam style: *short, self-contained derivations on named distributions*. Nothing is asked that needs more than one page of honest algebra. So the plan is built around **deriving, not reading**.

What the homeworks tell you about the exam

Fifteen problems across three sheets. Every single one has the form “here is a named family, derive the [UMVUE / Bayes estimate / minimax estimate / MP or UMP test].” There are **no** abstract measure-theoretic questions and **no** long asymptotic expansions. The families that recur: Binomial, Poisson, Exponential, Normal, Uniform, Cauchy, and power-series/negative-binomial. Master those seven and you have covered the observed question space.

Slot	Topic	What you actually do
Tonight (≈ 4 h) — rebuild the estimation core		
1.5 h	§1–§4: risk, unbiasedness, sufficiency, completeness, Rao–Blackwell, Lehmann–Scheffé	Read §1–§4 with a pen. Reproduce the factorisation theorem statement and the Lehmann–Scheffé proof from memory once.
1.0 h	§5–§7: exponential families, natural parameter space, ancillarity, Basu, Cramér–Rao	Write the exponential-family form for Bin, Poi, Exp, $N(\mu, \sigma^2)$, Gamma <i>from scratch</i> . Identify T , $c(\theta)$, natural space each time. Then learn the CRLB <i>attainment</i> condition in §7.
1.5 h	HW1 Q1–Q4, Q6	Attempt cold, 12 min each, then check the solutions PDF. These five are the highest-yield problems in the course.
Tomorrow morning (≈ 4 h) — Bayes and minimax (new)		
1.5 h	§8: prior/posterior, conjugacy, Bayes estimate = posterior mean, Bayes risk, improper priors	Derive all three conjugate pairs (Beta–Bin, Gamma–Poi, Normal–Normal) by hand. Do <i>not</i> memorise the posteriors; memorise the kernel-matching trick.
1.5 h	§9: minimax, constant-risk theorem, limits of Bayes rules	Reproduce the two-line proof of “Bayes + constant risk $\Rightarrow$ minimax” and the sequence version.
1.0 h	HW2 Q1–Q3	Cold attempt. Q1 and Q2 are the two canonical minimax arguments; expect one of them or a close cousin.
Tomorrow afternoon/evening (≈ 5 h) — testing (new, heaviest)		
2.0 h	§10: test functions, size/level/power, Neyman–Pearson with randomisation	Learn the NP statement exactly. Practise the loop: form $f_1/f_0 \rightarrow$ reduce to a statistic $\rightarrow$ invert to a region $\rightarrow$ set the size.
1.5 h	§11: MLR, Karlin–Rubin, UMP for one-sided hypotheses	Prove exponential family $\Rightarrow$ MLR. Know why two-sided UMP usually fails, and why $U[0, \theta]$ is the exception.
1.5 h	HW3 (all four) + HW2 Q5	These five are <i>exactly</i> the testing questions you will face. Do them under time.
Exam morning (≈ 2 h) — consolidate		
0.5 h	§12: consistency, b_n -consistency, MLE consistency	Light read; the cheapest section to score on.
1.0 h	§13 loose ends, then §14–§15: the cheat sheets	Skim §13, then re-derive every row of the distribution table from the exponential-family form. If you can rebuild the table, you are ready.
0.5 h	Skim your own notes 1–10	Only to catch anything idiosyncratic the professor stressed.

Triage — if you run short of time

Drop, in this order: the Cauchy/ancillarity fine print (§6), the simple-linear-regression consistency (§12), and the multiparameter completeness detail (§5). **Never** drop: Lehmann–Scheffé (§4), exponential-family completeness (§5), Neyman–Pearson (§10), Karlin–Rubin (§11). Those four carry the paper.

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1 Decision-theoretic setup (Lec 1)

$X \sim P_\theta$, $\theta \in \Theta$. An *estimator* $T = T(X)$ of $\psi(\theta)$ is judged by its *risk* under squared error:

$$R(\theta, T) = \mathbb{E}_\theta[(T - \psi(\theta))^2] = \text{MSE}_\theta(T) = \underbrace{(\mathbb{E}_\theta T - \psi(\theta))^2}_{\text{bias}^2} + \text{Var}_\theta(T).$$

- T is **unbiased** for ψ if $\mathbb{E}_\theta T = \psi(\theta)$ for *all* θ .
- T^* is **UMVUE** if it is unbiased and $\text{Var}_\theta(T^*) \leq \text{Var}_\theta(T)$ for every unbiased T and every θ .
- T is **inadmissible** if some T' has $R(\theta, T') \leq R(\theta, T)$ for all θ , with strict inequality somewhere.

The pitfall Lecture 1 opens with

There is *no* uniformly best estimator without restricting the class. For any fixed θ_0 , the silly estimator $T \equiv \theta_0$ has $R(\theta_0, T) = 0$, so no T^* can beat everything everywhere. That is exactly *why* we impose unbiasedness (giving UMVUE), or average the risk (giving Bayes), or look at the worst case (giving minimax). Be able to say this in one sentence — it is a standard opening bookwork part.

Uniqueness of the UMVUE (Lec 1) — a five-line “state and prove” answer

Claim. If a UMVUE exists, it is unique (a.s.).

Proof. Let T_1 and T_2 both be UMVUEs of $\psi(\theta)$, so both are unbiased and $\text{Var}_\theta(T_1) = \text{Var}_\theta(T_2) = \sigma^2$, say. Then $T = \frac{1}{2}(T_1 + T_2)$ is also unbiased, so by minimality of the UMVUE variance,

$$\sigma^2 \leq \text{Var}_\theta(T) = \frac{1}{4}[\text{Var}(T_1) + \text{Var}(T_2) + 2 \text{Cov}(T_1, T_2)] = \frac{\sigma^2}{4}[2 + 2\rho],$$

where $\rho = \text{Corr}_\theta(T_1, T_2)$. Hence $1 \leq \frac{1}{2}(1 + \rho)$, i.e. $\rho \geq 1$. But $\rho \leq 1$ always, so $\rho = 1$ — and correlation exactly 1 forces $T_2 = aT_1 + b$ a.s. with $a > 0$. Matching means and variances gives $a = 1$, $b = 0$, so $T_1 = T_2$ a.s. $\square$

This is a *separate* fact from Lehmann–Scheffé: it holds whenever a UMVUE exists, with no complete sufficient statistic in sight. Know both.

Worked example (Lec 1, and a classic exam part). $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$. Among estimators $T_c = c \sum_i (X_i - \bar{X})^2$ of σ^2 , minimise the MSE. Write $W = \sum (X_i - \bar{X})^2$, so $W/\sigma^2 \sim \chi_{n-1}^2$, $\mathbb{E}W = (n-1)\sigma^2$, $\text{Var } W = 2(n-1)\sigma^4$, hence $\mathbb{E}W^2 = \text{Var } W + (\mathbb{E}W)^2 = \sigma^4(n^2 - 1)$. Then

$$R(\sigma^2, T_c) = \mathbb{E}(cW - \sigma^2)^2 = c^2 \mathbb{E}W^2 - 2c\sigma^2 \mathbb{E}W + \sigma^4 = \sigma^4[c^2(n^2 - 1) - 2c(n - 1) + 1].$$

Minimising the quadratic in c gives $c^* = \frac{n-1}{n^2-1} = \frac{1}{n+1}$.

So $\frac{1}{n+1} \sum (X_i - \bar{X})^2$ beats the unbiased $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ in MSE at every σ^2 : **the UMVUE is inadmissible**. Memorise the constant $1/(n+1)$.

2 Does an unbiased estimator even exist? (Lec 2)

Before hunting for a UMVUE, ask whether ψ is *estimable*. The method is always the same: write $\mathbb{E}_\theta T = \psi(\theta)$ as an identity in θ , then use uniqueness of a series or transform expansion.

The three standard arguments

- (a) **Polynomial argument (Binomial).** $X \sim \text{Bin}(n, \theta)$, $\theta \in (0, 1)$. Then $\mathbb{E}_\theta T = \sum_{i=0}^n T(i) \binom{n}{i} \theta^i (1-\theta)^{n-i}$ is a *polynomial in θ of degree $\leq n$* . Hence ψ is unbiasedly estimable **iff** ψ is a polynomial of degree $\leq n$. In particular $\psi(\theta) = 1/\theta$ (or $(1-\theta)/\theta$) has *no* unbiased estimator: a degree- $\leq n$ polynomial cannot equal $1/\theta$ on an interval.
- (b) **Power-series argument (Poisson).** $X \sim \text{Poi}(\theta)$. Unbiasedness $\iff \sum_{i \geq 0} \frac{T(i)}{i!} \theta^i = \psi(\theta) e^\theta$ for all $\theta > 0$. So ψ is estimable **iff** $\psi(\theta) e^\theta$ is analytic on $(0, \infty)$, i.e. **iff** ψ is analytic; and then $T(i) = i! \times [i\text{-th Taylor coefficient of } \psi(\theta) e^\theta]$ is *unique*.
- (c) **Laplace-transform argument (Exponential).** $X \sim \frac{1}{\theta} e^{-x/\theta}$, $x > 0$. Then $\mathbb{E}_\theta T = \theta \iff \int_0^\infty T(x) e^{-x/\theta} dx = \theta^2$ for all $\theta > 0$. By uniqueness of the Laplace transform the solution T is unique (a.e.).

Remark. Whenever the unbiased estimator is *unique*, it is automatically the UMVUE — there is nothing to compare it with. This “uniqueness $\Rightarrow$ UMVUE” shortcut appears repeatedly and is worth one sentence in any answer.

Negative binomial (Lec 7). “Toss until 2 tails”: $f(x | \theta) = \binom{x-1}{1} \theta^{x-2} (1-\theta)^2$, $x \geq 2$. No unbiased estimator of θ exists (same polynomial-type obstruction). Contrast HW1 Q1, where θ is estimable for a differently parametrised negative binomial — estimability depends on the parametrisation, not on the distribution.

Cauchy (Lec 7). For $X \sim \frac{1}{\pi\{1+(x-\theta)^2\}}$ with $n = 1$ there is no unbiased estimator of θ ; X itself has no mean.

3 Sufficiency (Lec 2–3)

Definition 3.1. $T(X)$ is **sufficient** for θ if the conditional distribution of X given $T = t$ does not depend on θ .

Theorem 3.2 (Neyman–Fisher factorisation). T is sufficient $\iff$ the joint pdf/pmf factorises as

$$f(\mathbf{x} | \theta) = g(\theta, T(\mathbf{x})) \cdot h(\mathbf{x}),$$

with h free of θ and the θ -dependence entering only through T .

Model (n iid)	Sufficient statistic	Note
$\text{Bin}(m, \theta)$	$\sum X_i$	complete
$\text{Poi}(\theta)$	$\sum X_i$	complete; $\sum X_i \sim \text{Poi}(n\theta)$
$\text{Exp}(\text{mean } \theta)$	$\sum X_i$	complete; $\sum X_i \sim \text{Gamma}(n, \theta)$
$N(\theta, 1)$	$\bar{X}$	complete
$N(\mu, \sigma^2)$	$(\sum X_i, \sum X_i^2)$	complete (2-param. exp. family)
$U[0, \theta]$	$X_{(n)}$	complete
$U[\phi, \psi]$	$(X_{(1)}, X_{(n)})$	complete
$U[\theta, \theta + 1]$	$(X_{(1)}, X_{(n)})$	sufficient, not complete
$\text{Cauchy}(\theta)$	full order statistic	sufficient, not complete

Indicator bookkeeping for the uniform families. This is where marks are lost. For $U[\phi, \psi]$,

$$f(\mathbf{x} \mid \phi, \psi) = (\psi - \phi)^{-n} \prod_i \mathbb{I}[\phi < x_i < \psi] = (\psi - \phi)^{-n} \mathbb{I}[\phi < x_{(1)}] \cdot \mathbb{I}[x_{(n)} < \psi],$$

because “all $x_i > \phi$ ” $\iff$ “ $x_{(1)} > \phi$ ”. Always collapse the product of indicators into the two extreme order statistics before quoting the factorisation theorem.

Two-point parameter space (Lec 5). If $\Theta = \{\theta_0, \theta_1\}$, the likelihood ratio $T(x) = f_{\theta_1}(x)/f_{\theta_0}(x)$ is *sufficient*: take $h(x) = f_{\theta_0}(x)$ and $g(\theta, T) = T$ if $\theta = \theta_1$, $= 1$ if $\theta = \theta_0$. This is the bridge from estimation to testing — it is exactly why the Neyman–Pearson test depends on the data only through the likelihood ratio.

4 Completeness, Rao–Blackwell, Lehmann–Scheffé (Lec 2–3)

Definition 4.1. A sufficient statistic T is **complete** if for every function g ,

$$\mathbb{E}_\theta g(T) = 0 \quad \forall \theta \in \Theta \implies g(T) = 0 \text{ a.s. for all } \theta.$$

Why completeness is the right notion

If T is complete sufficient and $h_1(T), h_2(T)$ are both unbiased for ψ , then $\mathbb{E}_\theta[h_2(T) - h_1(T)] = 0$ for all θ , so $h_1 \equiv h_2$. There is at most one unbiased estimator that is a function of a complete sufficient statistic. Uniqueness is the whole point.

Theorem 4.2 (Rao–Blackwell). Let T be any unbiased estimator of $\psi(\theta)$ and S sufficient. Put $T^* = \mathbb{E}[T \mid S]$. Then T^* is a statistic (sufficiency makes it free of θ), $\mathbb{E}_\theta T^* = \psi(\theta)$, and $R(\theta, T^*) \leq R(\theta, T)$ for all θ .

Proof (memorise — it is three lines). $\mathbb{E}T^* = \mathbb{E} \mathbb{E}[T \mid S] = \mathbb{E}T = \psi(\theta)$, so the bias is unchanged. By the variance decomposition,

$$\text{Var}(T) = \mathbb{E}[\text{Var}(T \mid S)] + \text{Var}(\mathbb{E}[T \mid S]) \geq \text{Var}(T^*),$$

since the first term is ≥ 0 . Equal bias plus smaller variance gives smaller MSE. Equality holds iff $\text{Var}(T \mid S) = 0$ a.s., i.e. T is already a function of S . $\square$

Theorem 4.3 (Lehmann–Scheffé). If S is **complete** and sufficient and $h(S)$ is unbiased for $\psi(\theta)$, then $h(S)$ is the *UMVUE* of $\psi(\theta)$, and it is a.s. unique.

Proof. Let T be any unbiased estimator. Rao–Blackwellise: $T^* = \mathbb{E}[T \mid S]$ is unbiased, a function of S , and $\text{Var}(T^*) \leq \text{Var}(T)$. By completeness $T^* = h(S)$ a.s. Hence $\text{Var}(h(S)) \leq \text{Var}(T)$ for every unbiased T . $\square$

The UMVUE recipe — two routes, use whichever is shorter

Route A (guess-and-check). Find a complete sufficient S ; then find *any* function h with $\mathbb{E}_\theta h(S) = \psi(\theta)$, typically by expanding both sides in a series and matching coefficients. Done. (Used in HW1 Q1, Q2.)

Route B (Rao–Blackwellise). Find a crude unbiased estimator T_0 — often an indicator like $\mathbb{I}[X_1 > 0]$ or $\mathbb{I}[X_1 = 0]$ — and compute $\mathbb{E}[T_0 \mid S]$. The conditional distribution of X_1 given S is the only real work. (Used in HW1 Q3.)

Standard UMVUEs to know cold.

- $U[0, \theta]$: $X_{(n)}$ is complete sufficient with $\mathbb{E}X_{(n)} = \frac{n}{n+1}\theta$, so the UMVUE of θ is $\frac{n+1}{n}X_{(n)}$.
- $\text{Poi}(\theta)$, one observation: UMVUE of $e^{c\theta}$ is $(1+c)^X$; with n observations, $(1+c/n)^S$ where $S = \sum X_i$.
- $N(\theta, 1)$: UMVUE of $\Phi(\theta) = \mathbb{P}(X_1 > 0)$ is $\Phi(\bar{X}\sqrt{n/(n-1)})$.
- $\text{Bin}(n, \theta)$: UMVUE of θ is X/n ; UMVUE of $\theta(1-\theta)$ is $\frac{X(n-X)}{n(n-1)}$.

5 Exponential families (Lec 4–5)

Definition 5.1 (One-parameter). $f(x | \theta) = p(\theta) h(x) \exp\{c(\theta) T(x)\}$. For $X_1, \dots, X_n$ iid,

$$f(\mathbf{x} | \theta) = p(\theta)^n \left(\prod_i h(x_i) \right) \exp \left\{ c(\theta) \sum_i T(x_i) \right\},$$

so by the factorisation theorem $S = \sum_i T(X_i)$ is sufficient. Call $\eta = c(\theta)$ the **natural parameter** and $\Gamma = \{c(\theta) : \theta \in \Theta\}$ the **natural parameter space**.

Theorem 5.2 (Completeness criterion — the most used result in the course). *If the natural parameter space Γ contains an open interval (an interval with non-empty interior), then $S = \sum_i T(X_i)$ is **complete** sufficient.*

Multiparameter version (Lec 5): if $f(x | \theta) = p(\theta)h(x) \exp\{\sum_{j=1}^k c_j(\theta)T_j(x)\}$ with $\theta = (\theta_1, \dots, \theta_k)$, then $S = (\sum_i T_1(X_i), \dots, \sum_i T_k(X_i))$ is sufficient, and complete provided the natural parameter space $\{(c_1(\theta), \dots, c_k(\theta)) : \theta \in \Theta\}$ contains a k -dimensional rectangle with non-empty interior.

Family	$T(x)$	$c(\theta)$	natural space Γ	complete?
$\text{Bin}(m, \theta), \theta \in (0, 1)$	x	$\log \frac{\theta}{1-\theta}$	$\mathbb{R}$	yes
$\text{Poi}(\theta), \theta > 0$	x	$\log \theta$	$\mathbb{R}$	yes
$\text{Exp}(\text{mean } \theta)$	x	$-1/\theta$	$(-\infty, 0)$	yes
$N(\theta, 1)$	x	θ	$\mathbb{R}$	yes
$N(\mu, \sigma^2)$	(x, x^2)	$(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2})$	$\mathbb{R} \times (-\infty, 0)$	yes
$\text{Gamma}(\alpha, \beta)$	$(\log x, x)$	$(\alpha - 1, -1/\beta)$	open rectangle	yes
$U[0, \theta]$	not an exponential family (support depends on θ)			yes, shown directly
$\text{Cauchy}(\theta)$	not an exponential family			no

Where the criterion bites

$N(\theta, \theta^2)$ has natural parameters $(\frac{1}{\theta}, -\frac{1}{2\theta^2})$, which trace a *curve* in $\mathbb{R}^2$, not a rectangle. The criterion does not apply and completeness fails. Likewise $U[\theta, \theta + 1]$ is not an exponential family at all, and its sufficient statistic $(X_{(1)}, X_{(n)})$ is not complete because $X_{(n)} - X_{(1)}$ is ancillary. The phrase to write is: “the natural parameter space does not contain a rectangle with non-empty interior, so the criterion is unavailable.”

Power-series family (Lec 4).

$$f(x | \theta) = \frac{a_x \theta^x}{A(\theta)}, \quad x = 0, 1, 2, \dots, \quad A(\theta) = \sum_{x \geq 0} a_x \theta^x, \quad \theta \in (0, b), \quad a_x \geq 0.$$

This is an exponential family with $T(x) = x$, $c(\theta) = \log \theta$, natural space $(-\infty, \log b)$ — an open interval, so X (and $\sum X_i$) is complete sufficient. Two identities worth memorising, with $\psi(\theta) := \mathbb{E}_\theta X$:

$$\mathbb{E}_\theta X = \frac{\theta A'(\theta)}{A(\theta)} = \psi(\theta), \quad \text{Var}_\theta X = \theta \psi'(\theta).$$

Bernoulli, Binomial, Poisson, Geometric and Negative Binomial are all power-series families. (HW1 Q4 is exactly this; see the solutions PDF.)

6 Ancillarity and Basu's theorem (Lec 4)

Definition 6.1. $V(X)$ is *ancillary* if its distribution does not depend on θ .

Theorem 6.2 (Basu). If S is complete sufficient for θ and V is ancillary, then $S \perp\!\!\!\perp V$ under every P_θ .

How to spot an ancillary statistic in one line

Location family $f(x - \theta)$: any *difference* is ancillary — $X_i - X_j$, $X_{(n)} - X_{(1)}$, S^2 , the whole vector $(X_1 - \bar{X}, \dots, X_n - \bar{X})$.

Scale family $\frac{1}{\theta} f(x/\theta)$: any *ratio* is ancillary — X_i/X_j , $X_{(n)}/X_{(1)}$, $\sum_{j \leq p} X_j / \sum_{k \leq q} X_k$, $\bar{X}/S$.
Proof template: write $X_i = \theta Z_i$ (scale) or $X_i = \theta + Z_i$ (location) with Z_i free of θ ; the statistic is unchanged, hence its law is free of θ .

Cauchy: sufficiency without completeness. For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Cauchy}(\theta)$ the full order statistic $S = (X_{(1)}, \dots, X_{(n)})$ is sufficient (and minimal), but *not* complete: for bounded g , the statistic $g(X_{(i)} - X_{(j)})$ has a θ -free distribution, so with $c = \mathbb{E}g(X_{(i)} - X_{(j)})$ we get $\mathbb{E}_\theta[g(X_{(i)} - X_{(j)}) - c] = 0$ for all θ while $g - c \not\equiv 0$. This is the standard counterexample; know it.

The two textbook Basu applications.

- (i) $N(\mu, \sigma^2)$ with σ^2 known: $\bar{X}$ is complete sufficient for μ , and S^2 is ancillary, hence $\bar{X} \perp\!\!\!\perp S^2$. (The slick proof of the classical independence.)
- (ii) $\text{Exp}(\theta)$: $\sum X_i$ is complete sufficient, $\sum_{j \leq p} X_j / \sum_{k \leq q} X_k$ is scale-invariant hence ancillary, so the two are independent. (HW1 Q6.)

7 Information inequality / Cramér–Rao (Lec 2, HW1 Q4)

Under regularity (support free of θ , differentiation under the integral sign legal), the **Fisher information** in one observation is

$$I(\theta) = \text{Var}_\theta \left(\frac{\partial}{\partial \theta} \log f(X | \theta) \right) = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f \right)^2 \right] = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f \right],$$

and for n iid observations $I_n(\theta) = nI(\theta)$.

Theorem 7.1 (CRLB). For any T with $\mathbb{E}_\theta T = \psi(\theta)$, $\text{Var}_\theta(T) \geq \frac{[\psi'(\theta)]^2}{n I(\theta)}$.

Attainment — the exam-relevant half

Equality holds at every θ **iff** the score factorises as

$$\frac{\partial}{\partial \theta} \log f(\mathbf{x} | \theta) = k(\theta) [T(\mathbf{x}) - \psi(\theta)]$$

for some $k(\theta)$, i.e. **iff the model is a one-parameter exponential family and ψ is the mean of its natural sufficient statistic**. Consequences:

- $\bar{X}$ attains the CRLB for μ in $N(\mu, \sigma_0^2)$, for θ in $\text{Poi}(\theta)$, for θ in $\text{Exp}(\theta)$, and for $\mathbb{E}X$ in any power-series family. (The equality condition in Cauchy–Schwarz is the source.)
- The CRLB is *not* attained for e.g. σ^2 in $N(\mu, \sigma^2)$, or for non-linear ψ . A UMVUE can therefore exist with variance strictly above the CRLB — so “does not attain CRLB” never disproves UMVUE-hood.
- The CRLB is **meaningless** for $U[0, \theta]$: the support depends on θ , regularity fails, and indeed $\frac{n+1}{n}X_{(n)}$ has variance of order θ^2/n^2 , far below any bound you would naively compute.

Consistency of UMVUEs (Lec 4). If T_n is the UMVUE of $\psi(\theta)$ based on $X_1, \dots, X_n$ and there exists *some* unbiased sequence whose variance tends to 0, then $\text{Var}(T_n) \leq \text{Var}(\bar{T}_n) \rightarrow 0$, and with zero bias Chebyshev gives $T_n \xrightarrow{p} \psi(\theta)$.

8 Bayes estimation (Lec 6–7)

Prior $\pi(\theta)$ on Θ ; data $X \sim f(x | \theta)$. The **posterior** is

$$\pi(\theta | x) = \frac{f(x | \theta)\pi(\theta)}{\int_{\Theta} f(x | \theta')\pi(\theta') d\theta'} \propto f(x | \theta) \pi(\theta).$$

Theorem 8.1 (Bayes estimate under squared error). *The estimator minimising the **Bayes risk** $r(\pi, T) = \int_{\Theta} R(\theta, T)\pi(\theta) d\theta = \mathbb{E}_{\pi}[R(\theta, T)]$ is the **posterior mean** $T_{\pi}(x) = \mathbb{E}[\theta | X = x]$.*

Proof (know this). Write $r(\pi, T) = \iint (T(x) - \theta)^2 f(x | \theta)\pi(\theta) dx d\theta = \int m(x) \left[\int (T(x) - \theta)^2 \pi(\theta | x) d\theta \right] dx$, where m is the marginal density of X . For each fixed x the inner bracket is minimised over the number $T(x)$ by the mean of $\pi(\cdot | x)$, since $a \mapsto \mathbb{E}[(a - \theta)^2 | x]$ is minimised at $a = \mathbb{E}[\theta | x]$. The minimum value is the posterior variance, so $r(\pi, T_{\pi}) = \mathbb{E}_m[\text{Var}(\theta | X)]$ — a formula you will use repeatedly in the minimax problems. $\square$

Conjugate families — derive, do not memorise

Drop all factors free of θ and match the kernel.

Model	Prior	Posterior	Bayes estimate
$X \sim \text{Bin}(n, \theta)$	$\text{Beta}(p, q)$	$\text{Beta}(p + x, q + n - x)$	$\frac{p + X}{p + q + n}$
$X \sim \text{Poi}(\theta)$	$\text{Gamma}(\alpha, \lambda)$	$\text{Gamma}(\alpha + x, \frac{\lambda}{1 + \lambda})$	$\frac{\lambda(\alpha + X)}{1 + \lambda}$
$X \sim N(\theta, \sigma^2), \sigma^2$ known	$N(\mu, \tau^2)$	$N\left(\frac{\tau^2 x + \sigma^2 \mu}{\sigma^2 + \tau^2}, \frac{\sigma^2 \tau^2}{\sigma^2 + \tau^2}\right)$	$\frac{\tau^2 X + \sigma^2 \mu}{\sigma^2 + \tau^2}$
$X_{(n)}$ from $U[a, \theta]$	$\text{Pareto}(\alpha, a)$	$\text{Pareto}(\alpha + n, \max(a, x_{(n)}))$	$\frac{(\alpha + n) \max(a, X_{(n)})}{\alpha + n - 1}$

(Gamma parametrised by shape and *scale*.) For n iid observations replace x by the sufficient statistic: $\sum x_i$ for Bin/Poi, and $\bar{X}$ with $\sigma^2 \rightarrow \sigma^2/n$ for the Normal.

Three facts the professor emphasised.

1. **The posterior depends on the data only through a sufficient statistic.** If $f(x | \theta) = g(\theta, T(x))h(x)$ then $h(x)$ cancels between numerator and denominator: $\pi(\theta | x) = g(\theta, T(x))\pi(\theta) / \int g(\theta, T(x))\pi(\theta) d\theta$. Consequently *every Bayes and generalized Bayes estimate is a function of the sufficient statistic.*
2. **A Bayes estimate is essentially never unbiased.** Suppose T is both Bayes (so $T = \mathbb{E}[\theta | X]$) and unbiased (so $\mathbb{E}[T | \theta] = \theta$). Writing $\mathbb{E}_m$ for expectation under the joint law and conditioning two different ways,

$$\mathbb{E}_m[T\theta] = \mathbb{E}[\theta \mathbb{E}(T | \theta)] = \mathbb{E}[\theta^2], \quad \mathbb{E}_m[T\theta] = \mathbb{E}[T \mathbb{E}(\theta | X)] = \mathbb{E}[T^2].$$

Hence $\mathbb{E}_m[(T - \theta)^2] = \mathbb{E}[T^2] + \mathbb{E}[\theta^2] - 2\mathbb{E}_m[T\theta] = 0$, forcing $T = \theta$ a.s. — impossible except in degenerate cases. *Memorise this four-line argument.*

3. **Improper priors and generalized Bayes.** π need not integrate to 1. If the posterior is still a proper density, its mean is the *generalized Bayes estimate*. Examples: $X \sim N(\theta, 1)$ with $\pi \equiv 1$ gives $\pi(\theta | x) = N(x, 1)$ and estimate X ; with n iid observations, $\bar{X}$. For $U[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$ with $\pi \equiv 1$ the posterior is uniform on $[x_{(n)} - \frac{1}{2}, x_{(1)} + \frac{1}{2}]$ and the estimate is the midrange $\frac{1}{2}(X_{(1)} + X_{(n)})$ (HW2 Q3).

9 Minimax estimation (Lec 8–9)

Definition 9.1. T^* is **minimax** if $\sup_{\theta \in \Theta} R(\theta, T^*) = \inf_T \sup_{\theta \in \Theta} R(\theta, T)$. An estimator whose risk is constant in θ is called an **equalizer**.

Theorem 9.2 (Constant-risk criterion). If T_π is Bayes w.r.t. π and $R(\theta, T_\pi) = c$ is constant in θ , then T_π is minimax (and π is a least favourable prior).

Proof. For any T : $\sup_{\theta} R(\theta, T) \geq r(\pi, T) \geq r(\pi, T_\pi) = c = \sup_{\theta} R(\theta, T_\pi)$. The first step is “sup $\geq$ average”, the second is Bayes optimality. $\square$

Theorem 9.3 (Limit of Bayes rules — the workhorse). Let $\{\pi_k\}$ be priors with Bayes risks $r(\pi_k, T_{\pi_k}) \rightarrow c$. If T^* satisfies $\sup_{\theta} R(\theta, T^*) = c$, then T^* is minimax.

Proof. For any T and every k : $\sup_{\theta} R(\theta, T) \geq r(\pi_k, T) \geq r(\pi_k, T_{\pi_k})$. Let $k \rightarrow \infty$ to get $\sup_{\theta} R(\theta, T) \geq c = \sup_{\theta} R(\theta, T^*)$. $\square$

The three canonical minimax results

1. $X \sim \text{Bin}(n, \theta)$. With the prior $\text{Beta}(\frac{\sqrt{n}}{2}, \frac{\sqrt{n}}{2})$ the Bayes estimate is

$$T^* = \frac{X + \sqrt{n}/2}{n + \sqrt{n}},$$

whose risk is the *constant* $\frac{1}{4(1 + \sqrt{n})^2}$; hence T^* is minimax. By contrast the UMVUE

X/n has $\sup_{\theta} R = \frac{1}{4n}$, which is larger — **the UMVUE is not minimax.**

2. $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$. X is minimax with value 1. Use $\pi_k = N(0, k)$: the Bayes estimate is $kX/(k+1)$ with Bayes risk $\frac{k}{k+1} \rightarrow 1 = \sup_{\theta} R(\theta, X)$; apply the limit theorem. (HW2 Q1.)
3. $X \sim \text{Poi}(\lambda)$, $\lambda \in (0, \infty)$. *Every* estimator has infinite maximum risk. Use $\pi_k = \text{Gamma}(1, k)$, whose Bayes risk is $k^2/(1+k) \rightarrow \infty$. An unbounded parameter space with unbounded variance kills the problem. (HW2 Q2.) The same happens for $U[a, \theta]$ with θ unbounded.

10 Testing: setup and Neyman–Pearson (Lec 6, 9)

$H_0 : \theta \in \Theta_0$ vs $H_A : \theta \in \Theta_A$, $\Theta = \Theta_0 \cup \Theta_A$ disjointly.

Definition 10.1. A **test function** is $\phi : \mathcal{X} \rightarrow [0, 1]$; $\phi(x)$ is the probability of rejecting H_0 when $X = x$. Its **power function** is $\beta_\phi(\theta) = \mathbb{E}_\theta \phi(X)$. The **size** is $\sup_{\theta \in \Theta_0} \beta_\phi(\theta)$; ϕ has **level** α if its size is $\leq \alpha$. Type I error = $\beta_\phi(\theta)$ for $\theta \in \Theta_0$; Type II error = $1 - \beta_\phi(\theta)$ for $\theta \in \Theta_A$.

Sufficiency reduces the problem (Lec 6). If T is sufficient and ϕ is any test, then $\psi(T) := \mathbb{E}[\phi(X) | T]$ is a legitimate test function (free of θ by sufficiency) with $\mathbb{E}_\theta \psi = \mathbb{E}_\theta \phi$ for all θ — identical size and power. So it is enough to search among tests depending only on a sufficient statistic.

Theorem 10.2 (Neyman–Pearson lemma). Test $H_0 : X \sim f_0$ against $H_A : X \sim f_1$ (both simple). Fix $\alpha \in (0, 1)$.

(i) (**Existence**) There exist $k \geq 0$ and $\gamma \in [0, 1]$ such that

$$\phi^*(x) = \begin{cases} 1, & f_1(x) > k f_0(x), \\ \gamma, & f_1(x) = k f_0(x), \\ 0, & f_1(x) < k f_0(x), \end{cases} \quad \mathbb{E}_0 \phi^* = \alpha.$$

(ii) (**Sufficiency**) Any such ϕ^* is most powerful (MP) of its size.

(iii) (**Necessity**) Any MP level- α test has this form a.e., except possibly on $\{f_1 = k f_0\}$.

Proof of (ii) — short and often asked. Let ϕ be any test with $\mathbb{E}_0 \phi \leq \alpha = \mathbb{E}_0 \phi^*$. On $\{f_1 > k f_0\}$ we have $\phi^* = 1 \geq \phi$; on $\{f_1 < k f_0\}$ we have $\phi^* = 0 \leq \phi$. In both cases $(\phi^* - \phi)(f_1 - k f_0) \geq 0$, and this holds trivially on $\{f_1 = k f_0\}$ too. Integrating,

$$0 \leq \int (\phi^* - \phi)(f_1 - k f_0) = (\mathbb{E}_1 \phi^* - \mathbb{E}_1 \phi) - k(\mathbb{E}_0 \phi^* - \mathbb{E}_0 \phi) \leq \mathbb{E}_1 \phi^* - \mathbb{E}_1 \phi,$$

since $\mathbb{E}_0 \phi^* - \mathbb{E}_0 \phi \geq 0$ and $k \geq 0$. Hence $\mathbb{E}_1 \phi^* \geq \mathbb{E}_1 \phi$: ϕ^* is MP. □

The NP algorithm — follow it mechanically

1. Write $\Lambda(x) = f_1(x)/f_0(x)$ and **simplify until only a statistic remains**.
2. Determine the shape of $\{\Lambda > k\}$: is it a one-sided region in that statistic (usual), or an interval, or a union of two tails? (Λ need not be monotone — see HW2 Q5.)
3. Re-express the region in terms of a statistic with a *known null distribution*.
4. Solve $\mathbb{P}_0(\text{region}) = \alpha$ for the cutoff. If the null distribution is **discrete**, the exact size is generally unattainable, so **randomise**: take the smallest c with $\mathbb{P}_0(T > c) \leq \alpha$ and set $\gamma = \frac{\alpha - \mathbb{P}_0(T > c)}{\mathbb{P}_0(T = c)}$.
5. State the power if asked.

Support matters. If $f_1 > 0$ where $f_0 = 0$, the ratio is $+\infty$ there and the MP test *must* reject on that set — this costs nothing in size and gains power. That is the entire mechanism behind the $U[0, \theta]$ problem (HW3 Q4).

11 MLR and UMP tests (Lec 9)

Definition 11.1. $\{f_\theta : \theta \in (a, b)\}$ has **monotone likelihood ratio (MLR)** in $T(x)$ if for every $\theta_0 < \theta_1$ the ratio $f_{\theta_1}(x)/f_{\theta_0}(x)$ is a non-decreasing function of $T(x)$.

Theorem 11.2. *The one-parameter exponential family $f(x | \theta) = p(\theta)h(x)e^{c(\theta)T(x)}$ has MLR in $T(x)$ provided $c(\theta)$ is non-decreasing in θ .*

Proof. $\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = \frac{p(\theta_1)}{p(\theta_0)} \exp\{[c(\theta_1) - c(\theta_0)]T(x)\}$, and $c(\theta_1) - c(\theta_0) \geq 0$, so the ratio is a non-decreasing function of $T(x)$. $\square$

Theorem 11.3 (Karlin–Rubin). *If the family has MLR in T , then for $H_0 : \theta \leq \theta_0$ vs $H_A : \theta > \theta_0$ the test*

$$\phi^*(x) = \begin{cases} 1, & T(x) > c \\ \gamma, & T(x) = c \\ 0, & T(x) < c \end{cases} \quad \text{with } \mathbb{E}_{\theta_0} \phi^* = \alpha$$

is UMP of level α . Moreover $\beta_{\phi^}(\theta)$ is non-decreasing in θ , so the size is attained at $\theta = \theta_0$.*

Why this works — the sentence to write in the exam

Fix any $\theta_1 > \theta_0$. By Neyman–Pearson, the MP test of the *simple* $H_0 : \theta = \theta_0$ against $H_A : \theta = \theta_1$ rejects for large $f_{\theta_1}/f_{\theta_0}$, i.e. for large T by MLR. **The cutoff c depends only on α and the null distribution, not on θ_1** — so the same test is MP against every $\theta_1 > \theta_0$ simultaneously, hence UMP. The monotone power function then upgrades $H_0 : \theta = \theta_0$ to $H_0 : \theta \leq \theta_0$ without changing the size.

MLR status of the standard families. Binomial (in x), Poisson (in $\sum x_i$), Exponential (in $\sum x_i$), $N(\theta, 1)$ (in $\bar{X}$) and $U[0, \theta]$ (in $x_{(n)}$, in the generalised sense) all have MLR. **Cauchy does not:** $f_{\theta_1}/f_{\theta_0}$ is a ratio of two quadratics, which tends to 1 as $x \rightarrow \pm\infty$ in both directions, so it cannot be monotone.

Two-sided alternatives

For $H_0 : \theta = \theta_0$ vs $H_A : \theta \neq \theta_0$ in an MLR family, no UMP test exists: the best test against $\theta_1 > \theta_0$ rejects for large T , the best against $\theta_1 < \theta_0$ rejects for small T , and no single test does both. One then restricts to unbiased tests (UMPU). **Exception:** in non-regular families where the support moves with θ , a two-sided UMP test can exist — $U[0, \theta]$ with $H_0 : \theta = 1$ is the standard example (HW3 Q4), because rejecting on $\{X > 1\}$ is free under H_0 .

12 Consistency and asymptotics (Lec 4, 10)

Definition 12.1. T_n is (weakly) **consistent** for $\psi(\theta)$ if $T_n \xrightarrow{p} \psi(\theta)$ under P_θ for every θ . For $b_n \uparrow \infty$, T_n is **b_n -consistent** if $b_n(T_n - \theta) = O_p(1)$, i.e. $T_n - \theta = O_p(1/b_n)$.

Sufficient condition: if $\mathbb{E}_\theta T_n \rightarrow \psi(\theta)$ and $\text{Var}_\theta(T_n) \rightarrow 0$ then T_n is consistent (Chebyshev). This handles almost every exam instance.

Sample median (Lec 10). Let $F(\theta) = \frac{1}{2}$ with F strictly increasing and continuous near θ , $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$, and $\tilde{X}_n$ the sample median. Then

$$\mathbb{P}(\tilde{X}_n - \theta > \varepsilon) = \mathbb{P}(\text{at least half the } X_i > \theta + \varepsilon) = \mathbb{P}(\text{Bin}(n, 1 - F(\theta + \varepsilon)) \geq n/2) \rightarrow 0,$$

since $1 - F(\theta + \varepsilon) < \frac{1}{2}$; symmetrically for the other tail. Hence $\tilde{X}_n \xrightarrow{p} \theta$. *This is the right starting estimator for the Cauchy location problem, because $\bar{X}$ is not even consistent there — $\bar{X}$ is itself Cauchy(θ) for every n .*

Convergence in distribution (Lec 10). $X_n \xrightarrow{d} X$ — equivalently $F_n \xrightarrow{\mathcal{L}} F$ — means

$$F_n(x) \rightarrow F(x) \quad \text{only at the continuity points } x \text{ of } F.$$

The restriction is essential, not a technicality: take $X_n \equiv \frac{1}{n}$, so $F_n = \mathbb{I}[x \geq \frac{1}{n}]$, and $X \equiv 0$ with $F = \mathbb{I}[x \geq 0]$. Then $X_n \xrightarrow{d} X$, yet $F_n(0) = 0 \not\rightarrow 1 = F(0)$ — the convergence fails at the single discontinuity point $x = 0$. Note also that convergence in probability implies convergence in distribution, and the converse holds only when the limit is a constant.

MLE. $\hat{\theta}_n = \arg \max_{\theta} \prod_i f(X_i | \theta)$.

- Consistent for Binomial, Poisson, Exponential and Normal, and generally whenever the usual regularity conditions hold and the likelihood equation has a unique root for each n .
- $U[0, \theta]$: $\hat{\theta}_n = X_{(n)}$, with $\mathbb{E}X_{(n)} = \frac{n}{n+1}\theta$ (biased) but $\mathbb{P}(\hat{\theta}_n > \theta + \varepsilon) = 0$ and $\mathbb{P}(\hat{\theta}_n < \theta - \varepsilon) = (\frac{\theta - \varepsilon}{\theta})^n \rightarrow 0$, so consistent — and in fact n -consistent, not $\sqrt{n}$ -consistent.
- Cauchy: the likelihood equation $\sum_i \frac{2(x_i - \theta)}{1 + (x_i - \theta)^2} = 0$ is a polynomial equation with no closed form and possibly several roots; solve numerically starting from the sample median (HW2 Q4).

Simple linear regression (Lec 10). $Y_i = \alpha + \beta x_i + \varepsilon_i$, $\varepsilon_i \stackrel{\text{iid}}{\sim}$ with mean 0, variance σ^2 .

$$\hat{\beta}_{LS} = \frac{\sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_i (x_i - \bar{x})^2}, \quad \mathbb{E}\hat{\beta}_{LS} = \beta, \quad \text{Var}(\hat{\beta}_{LS}) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}.$$

So $\hat{\beta}_{LS}$ is consistent **iff** $\sum_i (x_i - \bar{x})^2 \rightarrow \infty$: the design points must be sufficiently spread out. And $\hat{\alpha}_{LS} = \bar{Y} - \hat{\beta}_{LS}\bar{x}$ is consistent provided additionally $\bar{x}$ converges to a constant.

13 Loose ends the professor covered

Short items that came up in class and are easy marks if they appear, but which do not belong to any of the threads above.

Method of scoring for the MLE (Lec 8). The likelihood equation is $\sum_i \frac{\partial}{\partial \theta} \log f(X_i | \theta) = 0$. Newton–Raphson would iterate using the observed second derivative $\sum_i \frac{\partial^2}{\partial \theta^2} \log f(X_i | \theta)$; the *method of scoring* replaces it by its expectation, $-nI(\theta)$, giving

$$\theta^{(m+1)} = \theta^{(m)} + \frac{1}{nI(\theta^{(m)})} \sum_i \frac{\partial}{\partial \theta} \log f(X_i | \theta) \Big|_{\theta^{(m)}}.$$

The point of the substitution: the Fisher information is often free of θ (e.g. for a location parameter), which makes the iteration cheaper and more stable.

Invariance (Lec 8). Let ψ be a function of θ .

- If $\hat{\theta}$ is *unbiased* for θ , then $\psi(\hat{\theta})$ is unbiased for $\psi(\theta)$ **only when ψ is linear** (by Jensen, a strictly convex or concave ψ introduces bias). The same caveat applies to Bayes estimates.
- The **MLE is invariant** under any one-one ψ : $\widehat{\psi(\theta)} = \psi(\hat{\theta})$. This is a genuine advantage of maximum likelihood over unbiasedness, and a standard one-mark comparison question.

Multinomial as a multiparameter exponential family (Lec 5, Ex 2). With $\mathbf{Y} = (Y_1, \dots, Y_k)$, $\sum_i Y_i = n$ and $\theta = (p_1, \dots, p_{k-1})$,

$$f(\mathbf{y}) = \frac{n!}{\underbrace{\prod_i y_i!}_h} \underbrace{(1 - p_1 - \dots - p_{k-1})^n}_{p(\theta)} \exp \left\{ \sum_{j=1}^{k-1} \underbrace{\log \frac{p_j}{1 - p_1 - \dots - p_{k-1}}}_{c_j(\theta)} \underbrace{y_j}_{T_j} \right\},$$

so $S = (Y_1, \dots, Y_{k-1})$ is complete sufficient (the natural parameter space is all of $\mathbb{R}^{k-1}$, which contains a rectangle).

Beta model (Lec 3, Ex 6). For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Beta}(\theta, \phi)$, factorisation gives the sufficient statistic $S = (\prod_i X_i, \prod_i (1 - X_i))$ — note it is a pair of *products*, not sums, because the exponential-family statistic here is $(\log x, \log(1 - x))$.

Zero functions (Lec 4). g is a *zero function* for the family $\{P_\theta\}$ if $\mathbb{E}_\theta g(X) = 0$ for every $\theta \in \Theta$. Completeness is exactly the statement that the only zero function is $g \equiv 0$; exhibiting a non-trivial zero function is therefore the standard way to *disprove* completeness.

Conjugate families in general (Lec 7). The class of *all* densities on Θ is trivially conjugate for any model, and is useless — a conjugate family earns its keep only by being *small*, so that updating is arithmetic on a few hyperparameters. The general construction $\pi(\theta) \propto \pi_0(\theta) \prod_{j=1}^k f(y_j | \theta)$ (“prior = base measure plus k imaginary observations”) is conjugate for every f ; it collapses to a two-parameter family precisely when f is an exponential family, which is why the Cauchy has no useful conjugate family.

Simulating a Cauchy (Lec 7). If $U \sim U[0, 1]$ then $X = \tan\{\pi(U - \frac{1}{2})\} + \theta \sim \text{Cauchy}(\theta)$ — inverse-cdf sampling, since the Cauchy cdf is $F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x - \theta)$.

14 Distribution cheat sheet

Model (n iid)	Complete suff. S	UMVUE of θ	Bayes (conj. prior)	UMP for $\theta \leq \theta_0$ vs $> \theta_0$
Bern(θ)	$\sum X_i$	$\bar{X}$	$\frac{p+S}{p+q+n}$ (Beta)	reject $S > c$, randomise
Poi(θ)	$\sum X_i$	$\bar{X}$	$\frac{\lambda(\alpha+S)}{1+n\lambda}$ (Gamma)	reject $S > c$, randomise
Exp(θ)	$\sum X_i$	$\bar{X}$	inverse-Gamma	reject $2S/\theta_0 > \chi_{2n, \alpha}^2$
$N(\theta, 1)$	$\bar{X}$	$\bar{X}$	$\frac{\tau^2 \bar{X} + (\sigma^2/n)\mu}{\sigma^2/n + \tau^2}$	reject $\bar{X} > \theta_0 + z_\alpha/\sqrt{n}$
$N(\mu, \sigma^2)$	$(\sum X_i, \sum X_i^2)$	$\bar{X}$ and S^2	—	t / χ^2 tests
$U[0, \theta]$	$X_{(n)}$	$\frac{n+1}{n} X_{(n)}$	Pareto	reject $X_{(n)} > \theta_0 \alpha^{1/n}$
Cauchy(θ)	order stat. (not complete)	none in closed form	—	no MLR, no UMP

Useful facts. For $\text{Exp}(\theta)$, $\sum_i X_i \sim \text{Gamma}(n, \theta)$ and $\frac{2}{\theta} \sum_i X_i \sim \chi_{2n}^2$ — the standard route to an exact cutoff. For $U[0, \theta]$, $X_{(n)}$ has density nx^{n-1}/θ^n on $[0, \theta]$, so $\mathbb{E}X_{(n)} = \frac{n\theta}{n+1}$ and $\text{Var} X_{(n)} = \frac{n\theta^2}{(n+1)^2(n+2)}$. For $\text{Laplace}(0, 1)$ with density $\frac{1}{2}e^{-|x|}$: $\mathbb{P}(|X| > t) = e^{-t}$ for $t \geq 0$. $\mathbb{E}[X_1 | \bar{X} = t] = t$ and $\text{Var}(X_1 | \bar{X} = t) = \sigma^2(n-1)/n$ in the normal model.

15 One-page decision tree

Read the question, then follow the arrow

“Find the UMVUE of $\psi(\theta)$ ”

→ Is the family exponential? Identify T , $c(\theta)$, and check the natural parameter space contains an open interval $\Rightarrow S = \sum T(X_i)$ is complete sufficient. Then either match a series (Route A) or Rao–Blackwellise an indicator (Route B). If the support depends on θ , use the extreme order statistic and verify completeness directly.

“Does an unbiased estimator exist?”

→ Set $\mathbb{E}_\theta T = \psi(\theta)$ and expand. Binomial $\Rightarrow$ polynomial of degree $\leq n$. Poisson $\Rightarrow \psi(\theta)e^\theta$ analytic. Exponential $\Rightarrow$ Laplace transform uniqueness.

“Does it attain the CRLB?”

→ Compute $\partial_\theta \log L$; if it equals $k(\theta)[T - \psi(\theta)]$ then yes, and $\text{Var}(T) = \psi'(\theta)/k(\theta)$. Otherwise no. Check regularity first — moving support means the CRLB does not apply at all.

“Show A and B are independent”

→ Basu. Show A is complete sufficient and B is location- or scale-invariant, hence ancillary. (Or, in the normal case, use joint normality and zero covariance.)

“Find the Bayes estimate”

→ Posterior $\propto$ likelihood $\times$ prior; recognise the kernel; report the posterior mean. If the prior is improper, check the posterior is proper and report the generalized Bayes estimate.

“Show T is minimax”

→ Constant risk? Find a prior making T Bayes and quote the constant-risk theorem. Otherwise build π_k with $r(\pi_k) \rightarrow \sup_\theta R(\theta, T)$ and quote the limit theorem. If Θ is unbounded and Bayes risks blow up, conclude every estimator has infinite maximum risk.

“Find the MP test” (both hypotheses simple)

→ Neyman–Pearson. Form f_1/f_0 , find the shape of $\{\Lambda > k\}$, set the size, randomise if the distribution is discrete or if Λ has ties.

“Find the UMP test” (one-sided)

→ Establish MLR (exponential family with monotone $c(\theta)$, or a direct computation), then quote Karlin–Rubin and set the cutoff from the null distribution at the boundary θ_0 .

“Find the UMP test” (two-sided)

→ Normally none exists; say why (the one-sided best tests point in opposite directions). Unless the support moves with θ , in which case rejecting on the impossible-under- H_0 region is free and a UMP test can exist.

Sources: lecture notes 1–10 (29 Jul — 28 Aug 2026) and Homeworks 1–3.

Companion document: *PI_Assignment_Solutions.pdf* — worked solutions to all 15 homework problems.